

Md Abu Sayed

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RESEARCH INTERESTS

Machine learning for prediction from incomplete, noisy, or weakly labeled observations: temporal and generative deep learning, uncertainty-aware prediction, simulation-based supervision, and interpretable decision support. Application domains: **cardiovascular AI and intravascular image analysis; intent recognition and trajectory prediction in multi-agent and autonomous systems**; human-robot collaboration; computer vision.

EDUCATION

University of Nevada, Reno

Reno, NV

Ph.D., Computer Science and Engineering

Aug 2021 – May 2026

- **Dissertation:** Deep Generative and Explainable Learning Frameworks for Intent Recognition in Naval Domain. Advisors: Prof. Monica Nicolescu and Prof. Mircea Nicolescu.

University of Nevada, Reno

Reno, NV

M.S., Computer Science and Engineering (earned en route to Ph.D.)

Aug 2021 – Dec 2023

- **Thesis:** ThreatMap: A Framework for Enhancing Security and Decision-Making for Naval Agents

Khulna University

Khulna, Bangladesh

B.S., Computer Science and Engineering

Dec 2014 – Jan 2019

- **Thesis:** Automated Method to Segment Retinal Blood Vessels from Color Fundus Photographs

RESEARCH EXPERIENCE

Postdoctoral Researcher, AI Initiatives

Jun 2026 – Present

Center for Digital Cardiovascular Innovations, Miller School of Medicine, Univ. of Miami

Miami, FL

- **Intravascular image segmentation.** Multi-class segmentation of lumen, vessel wall, and plaque composition (calcium, fibrous, fibrolipidic) across **HD-IVUS and NIRS-IVUS**, with upcoming extensions to angiography, OCT, and OCT-NIRS; trained on 40,000+ expert-labeled IVUS frames, with inter-observer variability as the reference level for validation.
- **Calcium scoring for stent outcome prediction.** Automated IVUS-based calcium scoring to predict stent under-expansion before intervention; OCT-based scoring upcoming.
- **Clinical decision support for PCI.** (i) Lesion-preparation strategy recommendation for calcified lesions (non-compliant balloon, intravascular lithotripsy, atherectomy) learned from a simulation-derived corpus spanning **1,200 lesion phenotypes and coronary anatomies**; (ii) bifurcation stenting technique recommendation.

Graduate Research Assistant

Aug 2021 – May 2026

Department of Computer Science and Engineering, University of Nevada, Reno

Reno, NV

- **Office of Naval Research (ONR)** maritime autonomy program (2021–25): intent recognition, trajectory prediction, and threat-aware decision support for naval agents under noisy and adversarial conditions.
- **MTITP** (dissertation capstone): multi-task Transformer with a three-stream encoder (kinematics, pairwise geometry, symbolic COLREGs features) that jointly predicts current intent, per-step future intent, and multi-modal trajectories; 99.2% seven-class intent accuracy, 99.4% future-intent mAP, and $2.3\times$ lower displacement error than the best Kalman filter (2.17 vs. 5.06 m); 98.4% vs. 88.0% for a published Bi-LSTM on noisy data.
- **MarITGAN:** intent-aware adversarial critic reducing trajectory error by 35% over a generic WGAN-GP critic with backbone and data fixed.
- Earlier intent classifiers (LSTM, Transformer) reached $\sim 97\%$ seven-class accuracy from the first 500 frames of an encounter.
- Lead developer of **NavySim** (IEEE CoG 2024; IEEE Transactions on Games 2026), a Unity multi-vessel simulator with real-time ML inference, and **ThreatMap**, a fused coverage/vulnerability/CPA risk visualization.
- Collaboration with **Huntington Ingalls Industries (HII)**: models were trained and evaluated on simulated encounter data provided by HII and developed for on-water deployment by HII as part of the ONR project.

AI Software Engineer Intern

Aug 2025 – May 2026

Pluto in Aquarius LLC

Reno, NV

- Architected **ICONIQ** (multimodal embedding-based recommendation, FastAPI, PostgreSQL/pgvector) and **AI-JINTSEE** (agentic LLM workflow orchestration with LangChain, RAG, multi-provider inference).
- Led a team of four students (2 M.S., 2 B.S.) on architecture, coding standards, and experiment design.

12 peer-reviewed publications (7 first-author; 2 journal articles), 5 submitted manuscripts.

Journal Articles

1. **Md Abu Sayed**, Korben DiArchangel, Mayamin Hamid Raha, Parvaneh Aliniya, Casey Brenner, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. NavySim 2.0: Enhanced multi-vessel simulation and analysis engine for advanced naval research. *IEEE Transactions on Games*, 2026
2. **Md Abu Sayed**, Sajib Saha, GM Atiqur Rahaman, Tanmai K Ghosh, and Yogesan Kanagasingam. An innovate approach for retinal blood vessel segmentation using mixture of supervised and unsupervised methods. *IET Image Processing*, 15(1):180–190, 2021

Peer-Reviewed Conference Papers and Book Chapters

3. **Md Abu Sayed**, Md Azizul Hakim, Ayesh Meepaganithage, Monica Nicolescu, and Mircea Nicolescu. Feature-aware deep learning for maritime intent recognition. In *2025 IEEE International Conference on Future Machine Learning and Data Science (FMLDS)*, pages 338–344. IEEE, 2025
4. **Md Abu Sayed**, Md. Azizul Hakim, Ayesh Meepaganithage, Tyler J Becker, Monica Nicolescu, and Mircea Nicolescu. Early classification of intentions for maritime domains using deep learning models. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, pages 1171–1178, 2025
5. Ayesh Meepaganithage, **Md Abu Sayed**, Mircea Nicolescu, and Monica Nicolescu. Proactive maritime threat prediction: Vessel intent classification with LSTMs and transformers using a sliding window approach. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE 2025)*, pages 3261–3267, 2025
6. Mayamin Hamid Raha, **Md Abu Sayed**, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. RAM: Resource allocation for multi-agent maritime environment. In Giuseppina Gini, Henk Nijmeijer, and Dimitar Filev, editors, *Informatics in Control, Automation and Robotics*, pages 69–95, Cham, 2026. Springer Nature Switzerland
7. **Md Abu Sayed**, Korben DiArchangel, Mayamin Hamid Raha, Parvaneh Aliniya, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. NavySim: A multi-vessel simulation and analysis engine for naval domains. In *2024 IEEE Conference on Games (CoG 2024)*, pages 1–8, 2024
8. **Md Abu Sayed**, Mayamin Hamid Raha, Parvaneh Aliniya, Monica Nicolescu, and Mircea Nicolescu. ThreatMap: A framework for enhancing security awareness and decision-making for naval agents. In *Proceedings of the 26th International Conference on Harbor, Maritime and Multimodal Logistic Modeling & Simulation (HMS 2024)*. CAL-TEK, 2024
9. Mayamin Hamid Raha, **Md Abu Sayed**, Sergiu Dascalu, Monica Nicolescu, and Mircea Nicolescu. Keep sailing: An investigation of effective navigation controls and subconscious learning in simulated maritime environment. In *International Conference on Information Technology-New Generations*, pages 313–322. Springer, 2024
10. Mayamin Raha, **Md Abu Sayed**, Monica Nicolescu, Mircea Nicolescu, and Sushil Louis. Maritime dynamic resource allocation and risk minimization using visual analytics and elitist multi-objective optimization. In *Proceedings of the 20th International Conference on Informatics in Control, Automation and Robotics - Volume 1: ICINCO*, pages 54–63. INSTICC, SciTePress, 2023
11. **Md Abu Sayed**, Sajib Saha, GM Rahaman, Tanmai K Ghosh, and Yogesan Kanagasingam. A semi-supervised approach to segment retinal blood vessels in color fundus photographs. In *Conference on Artificial Intelligence in Medicine in Europe*, pages 347–351. Springer, 2019
12. Tanmai K Ghosh, Sajib Saha, GM Rahaman, **Md Abu Sayed**, and Yogesan Kanagasingam. Retinal blood vessel segmentation: A semi-supervised approach. In *Iberian Conference on Pattern Recognition and Image Analysis*, pages 98–107. Springer, 2019

Submitted Manuscripts

13. Wei Wu, Evangelia I. Zacharaki, **Md Abu Sayed**, Sree Sindhu Vijayarao, Yash Vardhan Trivedi, Kaiming Li, Parth Munjal, Usama Oguz, Vineeth Dasari, Abdulkader Shaar, Sofoklis Chatzizisis, Akshat Banga, and Yiannis S. Chatzizisis. Deep learning model for multi-class segmentation of high-definition intravascular ultrasound. *Scientific Reports*, under review, 2026
14. S. Zhao, W. Wu, C. Lee, Y. Issa, H. Zeen Alden, S. Thapa, R. A. Pecora, FNU Nisha, S. S. Vijayarao, V. Miriyala, P. V. Singh, K. Goswami, M. F. Alam, **Md Abu Sayed**, E. S. Brilakis, G. Dangas, T. W. Johnson, G. Stankovic, Y. Murasato, J. F. Lassen, and Y. S. Chatzizisis. Head-to-head comparison of coronary artery bifurcation stenting strategies: A virtual clinical study. *JACC: Cardiovascular Interventions*, under review, 2026
15. Wei Wu, Sree Sindhu Vijayarao, Yaman Issa, Nikolaos Spiliadis, Iliana S. Hurtado Rendon, Shijia Zhao, Changkye Lee, **Md Abu Sayed**, Thomas W. Johnson, and Yiannis S. Chatzizisis. Head-to-head comparison of latest intracoronary imaging modalities. Research Letter submitted to *JACC: Cardiovascular Interventions*, 2026
16. **Md Abu Sayed**, Monica Nicolescu, and Mircea Nicolescu. Joint intent recognition and trajectory prediction for maritime vessel encounters. Submitted to *IEEE Transactions on Intelligent Transportation Systems*, 2026

17. **Md Abu Sayed**, Monica Nicolescu, and Mircea Nicolescu. Intent-aware critic design for adversarial trajectory generation in maritime encounters. Submitted to *Engineering Applications of Artificial Intelligence*, 2026

CONFERENCE PRESENTATIONS

- “Feature-aware deep learning for maritime intent recognition.” **IEEE FMLDS 2025**, Los Angeles, CA, USA.
- “NavySim: A multi-vessel simulation and analysis engine for naval domains.” **IEEE CoG 2024**, Milan, Italy.
- “ThreatMap: A framework for enhancing security awareness and decision-making for naval agents.” **HMS 2024**, Tenerife, Spain.

TEACHING EXPERIENCE

Lecturer (Instructor of Record) <i>Department of Computer Science and Engineering, The Millennium University</i>	Jun 2019 – Jul 2021 <i>Dhaka, Bangladesh</i>
– Independently taught Structured Programming, Data Structures, Database Systems, Digital Logic Design, Object-Oriented Analysis and Design, Software Engineering, and Artificial Intelligence & Neural Networks.	
– Designed syllabi, lectures, assignments, and examinations aligned with learning outcomes; supervised undergraduate capstone and term projects.	
– Course Coordinator: course scheduling and curriculum alignment across sections.	
Graduate Teaching Assistant <i>Department of Computer Science and Engineering, University of Nevada, Reno</i>	Aug 2021 – May 2026 <i>Reno, NV</i>
– CS 477/677 Analysis of Algorithms (4 semesters); CS 422/622 Introduction to Machine Learning; CS 491/691 Large Language Models and Multimodal AI; lab instructor, Programming for Data Science.	
– Led review sessions and weekly office hours; designed and graded homework, examinations, and programming assignments.	

MENTORING AND STUDENT SUPERVISION

- **Thesis supervisor**, Khulna University, Bangladesh, 2026–present. Projects I initiated and lead; co-supervised with Prof. G. M. Atiqur Rahaman:
 - M.S. thesis: pedestrian intent recognition for autonomous vehicles.
 - B.S. thesis: coronary OCT segmentation.
- **Technical lead**, Pluto in Aquarius LLC, 2025–26: supervised four students (2 M.S., 2 B.S.) on system architecture and experiment design.
- **Capstone supervisor**, The Millennium University, 2019–21: undergraduate software and AI projects.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, MATLAB

ML / CV: PyTorch, TensorFlow, Keras, OpenCV; segmentation, sequence models, generative models

Agentic AI & LLMs: agentic workflow orchestration (LangChain), multi-provider LLM integration, retrieval-augmented generation (RAG, multimodal RAG), embedding-based vector search

Systems: FastAPI, PostgreSQL/pgvector, MySQL, Git

Simulation: Unity, shader programming

SERVICE AND LEADERSHIP

- **Peer reviewer** (about 10 manuscripts): IEEE Conference on Games, *PLOS ONE*, *Information Systems*, and others.
- **Graduate Student Association, UNR** – Council Member; Chair, Awards Committee (2024–25).
- **Google Developer Group on Campus, UNR** – Co-Lead; co-organized DevFest Reno (2024–25).
- **International Students Club, UNR** – Vice President; organized Night of All Nations, 600 attendees (2023–24).